

AgrilIntelligence: An End-to-End Crop, Weather and Market Analytics & Prediction Platform

Abstract:

Agriculture is highly influenced by multiple interconnected factors such as weather conditions, soil characteristics, crop selection, cultivation practices, and market prices. However, these factors are often available across different datasets and platforms, making it difficult to obtain meaningful insights and make data-driven agricultural decisions. AgrilIntelligence is an end-to-end agricultural analytics and prediction platform designed to integrate agricultural, weather, and market datasets to provide meaningful insights and support crop yield prediction.

The project will collect and integrate data from sources such as data.gov.in agricultural datasets, weather data, and agricultural market price datasets. The data will undergo preprocessing, cleaning, missing-value handling, exploratory data analysis, statistical analysis, and feature engineering. Machine learning algorithms will then be applied to predict crop yield based on factors such as rainfall, temperature, soil quality, fertilizer usage, cultivated area, crop type, and region. Multiple models will be evaluated using appropriate performance metrics, and the best-performing model will be selected for deployment. An interactive dashboard will provide visualizations of crop trends, weather patterns, market prices, regional comparisons, and model-based yield predictions.

From a Data Science perspective, the project demonstrates the complete data analytics lifecycle, from multi-source data acquisition and preprocessing to exploratory analysis, machine learning, model evaluation, and visualization. It provides an opportunity to analyze relationships between environmental and agricultural factors and develop predictive insights that can support better agricultural planning.

The project also provides strong justification for its integration with Adaptive Software Engineering. Rather than being limited to a standalone machine learning model, AgrilIntelligence will be developed as a complete software system using an iterative and Agile approach. The platform can incorporate software engineering practices such as requirements analysis, user stories, Scrum-based development, Git version control, automated testing, CI/CD, Docker containerization, Kubernetes deployment, security testing, and cloud deployment. Furthermore, the machine learning lifecycle can be extended using MLOps practices such as data and model versioning, experiment tracking, model deployment, and application monitoring.

Thus, AgrilIntelligence serves as a common project for both Data Science and Adaptive Software Engineering, combining data-driven intelligence with modern software development

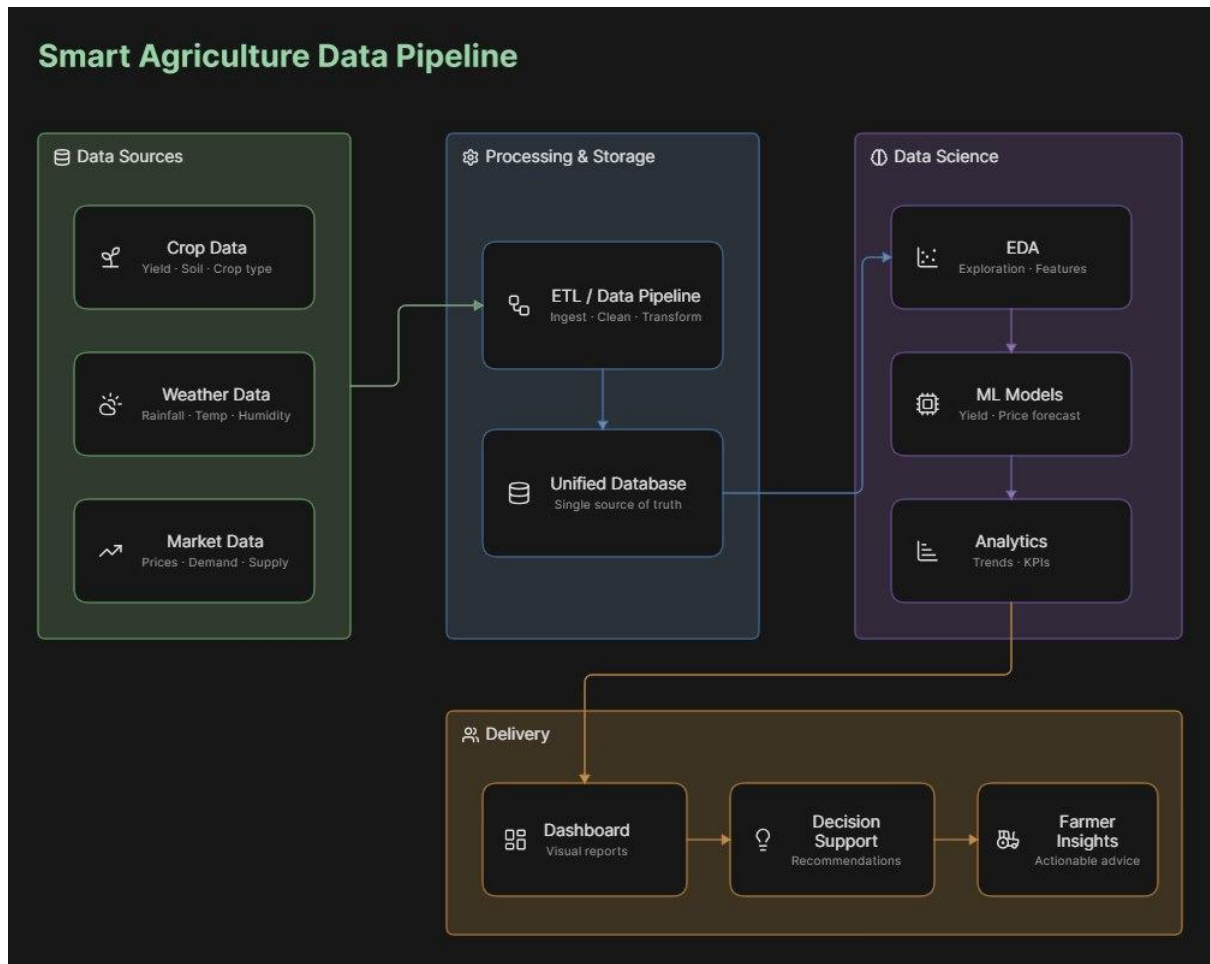
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and deployment practices. The project demonstrates how machine learning can be transformed from an isolated analytical model into a scalable, secure, maintainable, and deployable real-world application.



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